

Lesson 18 ROS2 Gazebo 3D Physics Simulation Platform

1. Gazebo Introduction

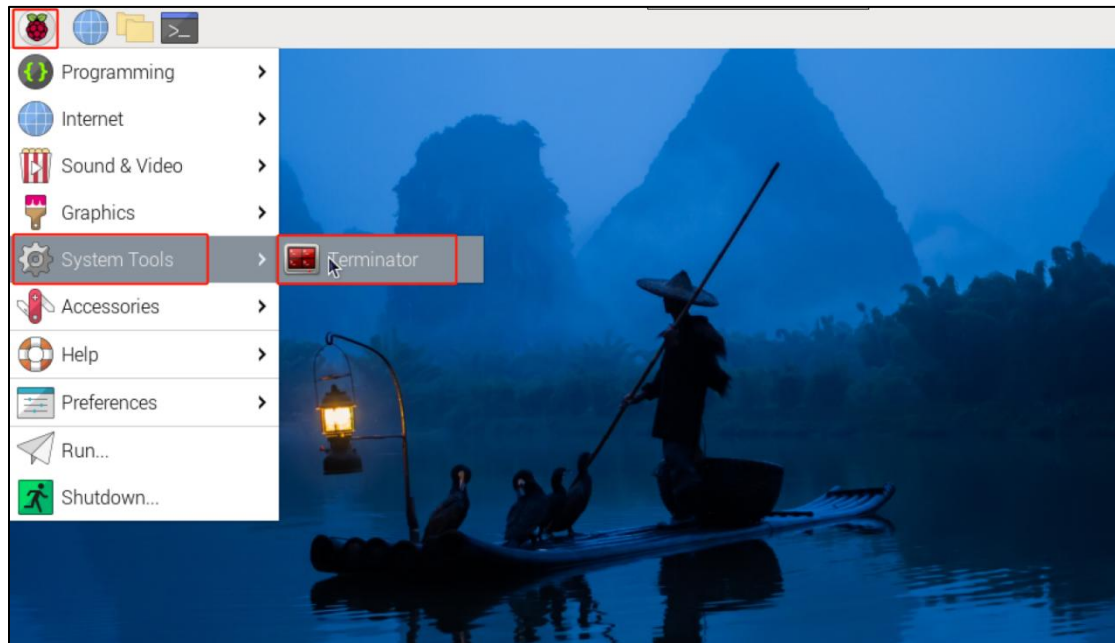
Gazebo is the most commonly used 3D physics simulation platform in ROS system. It supports dynamics engine, enabling high-quality graphic rendering. Not only can simulate robots and surrounding environments, but it can also incorporate physical properties such as friction and elasticity coefficients.

For instance, if we plan to develop a Mars robot car, the Martian surface environment can be simulated in Gazebo. Similarly, if we face constraints in conducting frequent experiences due to drone endurance and flight restrictions, we can utilize Gazebo for simulation. This allows us to develop algorithms in simulation first and then deploy them onto physical hardware once they are sufficiently refined.

Therefore, the similar simulation platforms like Gazebo can assist us in validating robot algorithms, optimizing robot designs, and testing robot scenarios and applications, which explores more possibilities for robot development.

2. Run Gazebo

- 1) Click  to select “**System Tools** → **Terminator**” in sequence.



2) Input the command **“ros2 launch ros_gz_sim_demos robot_description_publisher.launch.py”** and press Enter to run the **“robot_description_publisher.launch.py”** file.

```
ubuntu@raspberrypi:~$ ros2 launch ros_gz_sim_demos robot_description_publisher.launch.py
```

After successful execution, the simulation interface of Ignition and the PC software of Rviz will open, allowing us to visualize the data of the small ball.

